Cover Letter

10 September 2026

The Editor *Environment and Planning B: Urban Analytics and City Science* SAGE Publications

Dear Editor,

I am pleased to submit my manuscript, "Ghost Infrastructure: Historical Industrial Geography and the Persistence of Path Dependent Accessibility in Bochum, Germany," for consideration as an Original Manuscript in *Environment and Planning B: Urban Analytics and City Science*.

This study tests whether Bochum's 19th- and 20th-century coal-mining industrial geography, its coal mines and worker-housing colonies (Zechensiedlungen), still shapes present-day walking accessibility to essential services, more than 50 years after the region's last coal mine closed. Using the city's full pedestrian street network (69,393 nodes) and true network-distance accessibility modelling, the study finds a reversed effect from what was originally hypothesized: areas with lower present-day accessibility sit significantly farther from historical industrial sites than areas with higher accessibility (Welch's $t=42.887$, $p<0.00001$, Cohen's $d=0.589$), a pattern that holds after controlling for the most obvious confound, distance to the city center, and is corroborated by an independent Local Moran's I spatial-clustering analysis. The finding is tested for robustness across three walking-time thresholds and independently replicated in a second Ruhr Valley city, Essen, with the replication's partial (rather than complete) success reported in full.

I believe this manuscript is a good fit for *Environment and Planning B* because it applies quantitative, computational, and geospatial methods (network-distance accessibility modelling, spatial statistics, logistic regression, and a network-block bootstrap robustness check) to a substantive question about the spatial and morphological structure of a city, directly within the journal's stated scope. The manuscript also reports a partially non-replicating result (the confound-independence finding does not hold in Essen) and a hypothesis that the data does not support, in the interest of transparent and complete reporting rather than a tidier narrative.

This manuscript is original work, has not been published previously, and is not currently under consideration at any other journal. An earlier version was posted as a preprint on EarthArXiv (DOI: https://doi.org/10.31223/X5N501, posted 5 September 2026); it has not undergone peer review. The underlying code and data are openly available in a public repository, as described in the Data Availability Statement on the title page.

Generative AI (Claude, Anthropic) was used as an AI-assisted tool to inspect the GitHub repository for potential inconsistencies or contradictions between the documentation and the underlying code. The identified issues were subsequently reviewed and verified by the author, who independently addressed and resolved the bugs wherever possible. In cases where additional assistance was required, AI was used solely as a guidance and troubleshooting aid. Claude was also used to review and refine the manuscript for grammar, spelling, punctuation, and overall linguistic clarity.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

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